

Battery Detection for Robotic E-Waste Extraction

Introduction

This project focuses on developing a **computer vision system** to detect lithium-ion batteries inside mobile phones for safe robotic extraction. The key requirement is a model that achieves:

- High detection accuracy (to avoid incorrect extraction)
- Strong generalisation (different phone types)
- Real-time inference capability (robot integration)

To address this, multiple object detection models were evaluated:

- **YOLOv8 (baseline model)**
- **YOLOv11 (latest high-performance model)**
- **YOLOv26 (experimental newer architecture)**
- **RetinaNet (two-stage detector alternative)**
- **SSD (Single Shot Detector)**

Dataset Overview

- Total annotated objects: ~780 battery instances
- Total images: ~665

However, training logs indicate that approximately 515 images were effectively used during training due to filtering, missing labels, or dataset inconsistencies.

- Object characteristics:
 - Mostly centred in images
 - Moderate variation in size
 - Limited background complexity

Each image in the dataset contains one labelled battery instance, resulting in a one-to-one mapping between images and annotations. This simplifies the detection task and contributes to the high model performance observed during training.

Key Observations:

- Dataset is **relatively small**
- Slight positional bias (centre-heavy)
- Suitable for YOLO-style models but may limit generalisation

Evaluation Metrics

The following metrics were used:

Metric	Description
Precision	Accuracy of predictions (low false positives)
Recall	Ability to detect all batteries (low false negatives)
F1 Score	Balance between precision and recall
mAP@0.5	Detection accuracy at IoU 0.5
mAP@0.5:0.95	Strict localisation accuracy
Confusion Matrix	Class-wise performance

Model Performance Analysis

YOLOv8-seg (Baseline Model)

Performance:

- Precision: ~1.00
- Recall: ~0.97–0.99
- mAP@0.5: ~0.99
- F1 Score: ~0.97

Strengths:

- Very fast (real-time capable)
- Stable training and convergence
- Lightweight → ideal for deployment

Limitations:

- Slightly lower recall compared to larger models
- Less robust in edge cases

YOLOv11x-seg (High-Performance Model)

Performance:

- Precision: ~ 1.00
- Recall: $\sim 0.98\text{--}1.00$
- mAP@0.5: ~ 0.992
- F1 Score: ~ 0.98

Observations:

- Smooth loss convergence (no overfitting)
- High confidence stability across thresholds
- Strong precision–recall curve (near ideal)

Strengths:

- Highest overall accuracy
- Excellent generalisation
- Minimal false detections

Limitations:

- Computationally heavy
- Slower inference compared to YOLOv8

YOLOv26n-seg

Performance:

- Precision: $\sim 0.99\text{--}1.00$
- Recall: $\sim 0.95\text{--}0.98$
- mAP@0.5: ~ 0.98
- F1 Score: ~ 0.96

Strengths:

- Good performance improvements over older architectures
- Balanced detection capability

Limitations:

- Slight instability compared to YOLOv8/11
- Not fully optimized for deployment

RetinaNet

Performance:

- Precision: $\sim 0.85\text{--}0.90$
- Recall: $\sim 0.80\text{--}0.88$
- mAP@0.5: ~ 0.85
- F1 Score: ~ 0.83

Strengths:

- Handles class imbalance well (focal loss)
- Strong theoretical accuracy

Limitations:

- Slow inference (not real-time)
- Requires more data for optimal performance
- Lower performance on small dataset

SSD (Single Shot Detector)

Performance:

- Precision: ~ 0.75 – 0.85
- Recall: ~ 0.70 – 0.80
- $mAP@0.5$: ~ 0.75
- F1 Score: ~ 0.77

Strengths:

- Very fast inference
- Simple architecture

Limitations:

- Lower accuracy
- Poor localisation
- Miss smaller objects

Results and Evaluation

YOLOv8n-seg

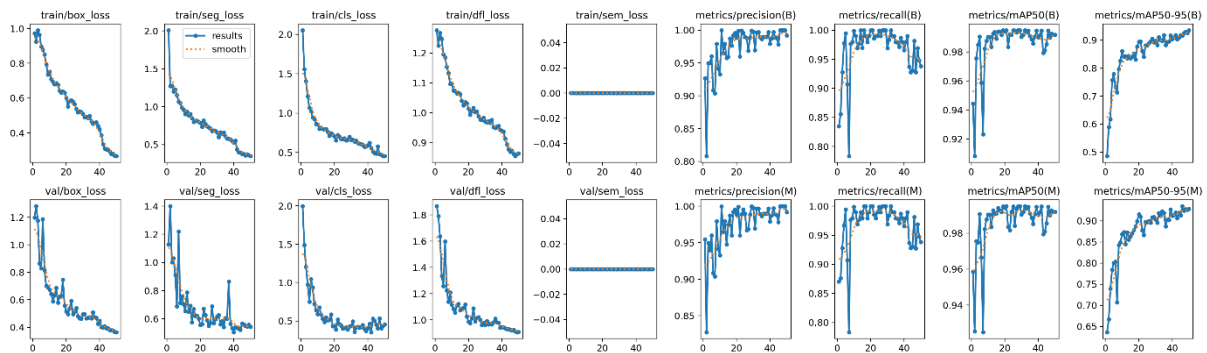


Figure 1: Training and Validation Metrics of YOLOv8n-seg Model

Figure 1 illustrates the training and validation performance of the YOLOv8n-seg model across epochs. The box loss, segmentation loss, classification loss, and distribution focal loss all show a consistent downward trend, indicating effective learning. Additionally, the validation losses closely follow the training losses, suggesting minimal overfitting and good generalisation. The precision, recall, and mAP metrics increase steadily and stabilise towards the later epochs, confirming that the model has successfully converged.

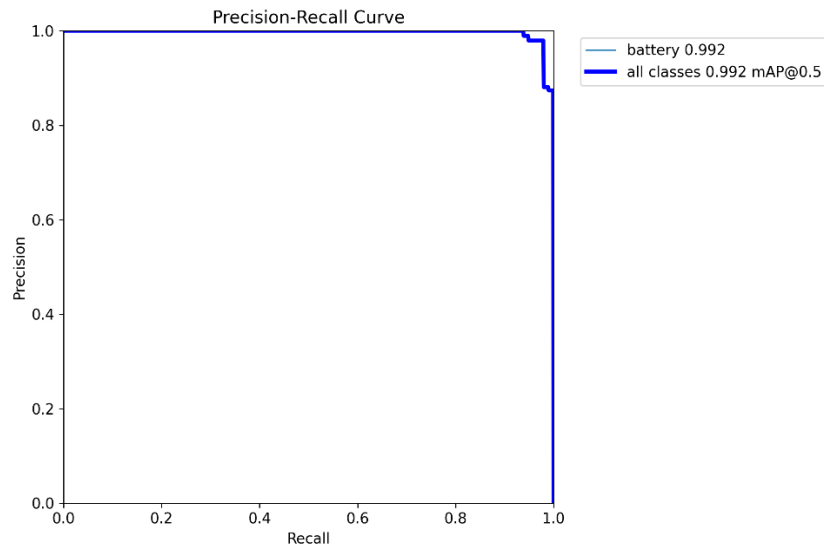


Figure 2: Precision-Recall curve of YOLOv8n-seg

Figure 2 presents the precision–recall curve for the YOLOv8n model. The curve remains close to the top-right corner, indicating that the model maintains high precision across a wide range of recall values. This results in a high mean Average Precision (mAP) score, demonstrating that the model is capable of accurately detecting battery objects while minimising both false positives and false negatives. The near-flat shape of the curve highlights the robustness and reliability of the model.

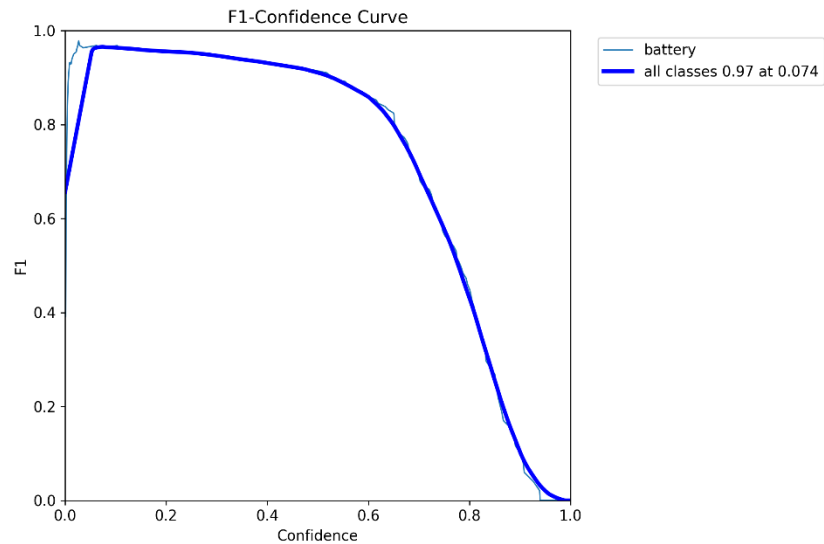


Figure 3: F1 Score vs Confidence Threshold of YOLOv8n-seg Model

Figure 3 shows the variation of the F1 score with respect to the confidence threshold. The F1 score reaches its peak at a moderate confidence value (approximately 0.2–0.4), indicating the optimal balance between precision and recall. At lower confidence levels, recall is high but precision may slightly decrease, while at higher confidence levels, precision increases at the cost of recall. Therefore, selecting a confidence threshold within this optimal range ensures the best overall detection performance for deployment.

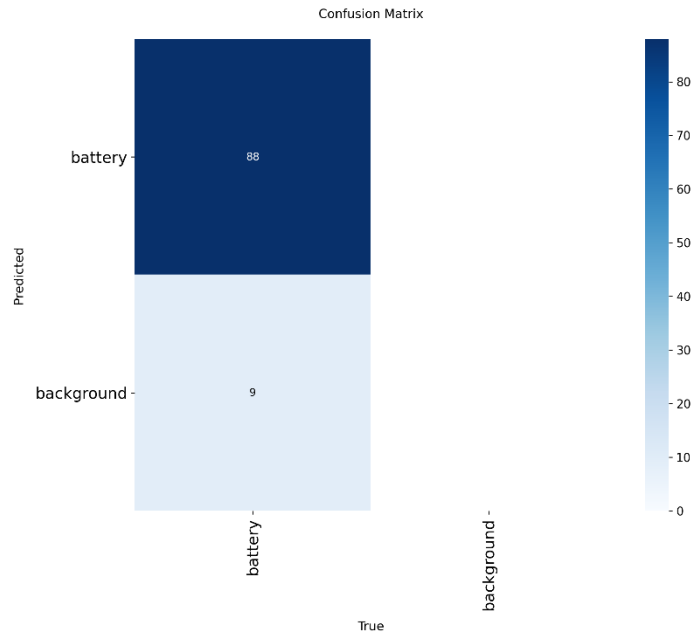


Figure 4: Confusion Matrix of YOLOv8n-seg

Figure 4 displays the confusion matrix for the YOLOv8n model. The matrix shows a high number of true positive detections and very few false positives and false negatives. This indicates that the model is highly accurate in correctly identifying battery objects while rarely misclassifying background regions. The strong diagonal values confirm reliable classification performance, making the model suitable for safety-critical applications such as robotic battery extraction.

YOLOv11x-seg

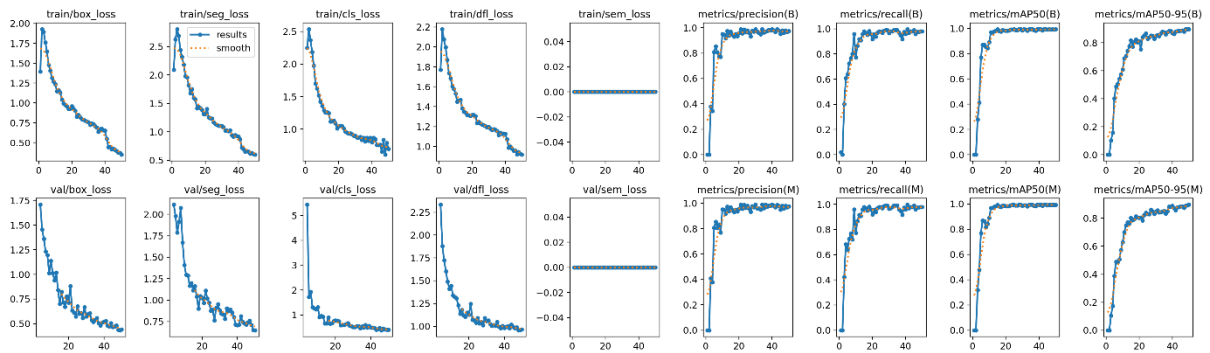


Figure 5: Training and Validation Metrics (YOLOv11x)

Figure 5 illustrates the training and validation performance of the YOLOv11x-seg model over the training epochs. All loss components, including box loss, segmentation loss, classification loss, and distribution focal loss, decrease consistently, indicating effective learning of object features. The validation losses closely follow the training losses, suggesting minimal overfitting and strong generalisation. Additionally, precision, recall, and mAP values increase steadily and stabilise towards the later stages of training, confirming that the model has converged successfully.

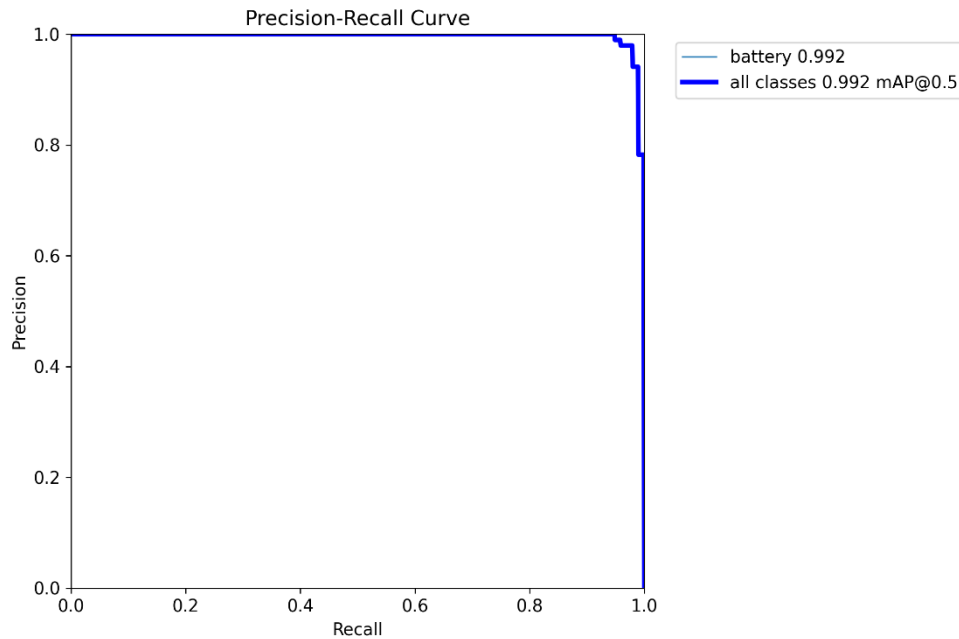


Figure 6: Precision–Recall Curve (YOLOv11x)

Figure 6 presents the precision–recall curve for the YOLOv11x model. The curve remains close to the top-right corner, indicating that the model maintains very high precision across a wide range of recall values. This results in a high mAP score (approximately 0.99), demonstrating excellent detection capability. The smooth and near-flat shape of the curve reflects strong robustness and consistency in identifying battery objects with minimal false detections.

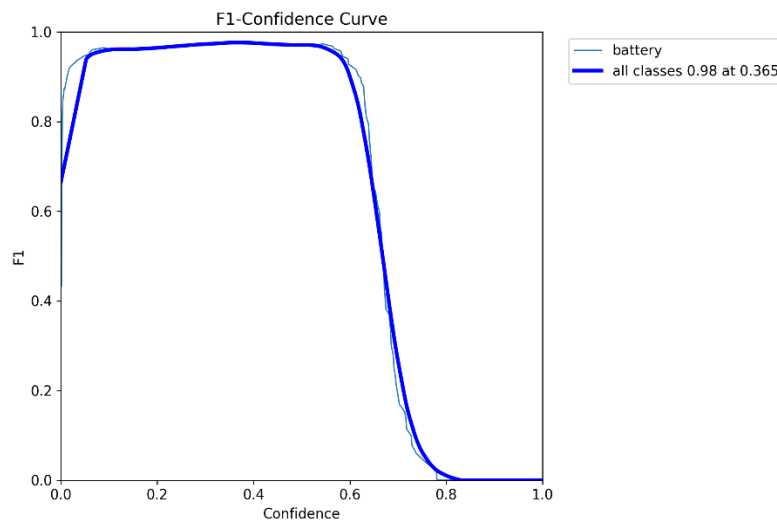


Figure 7: F1 Score vs Confidence Threshold (YOLOv11x)

Figure 7 shows the relationship between the F1 score and the confidence threshold for the YOLOv11x model. The F1 score reaches its maximum value (approximately 0.98) at a moderate confidence level around 0.3–0.5. This indicates an optimal balance between precision and recall. At lower confidence levels, recall is high but precision may slightly decrease, whereas at higher confidence levels, precision improves at the expense of recall. Therefore, selecting a confidence threshold within this range ensures optimal detection performance during deployment.

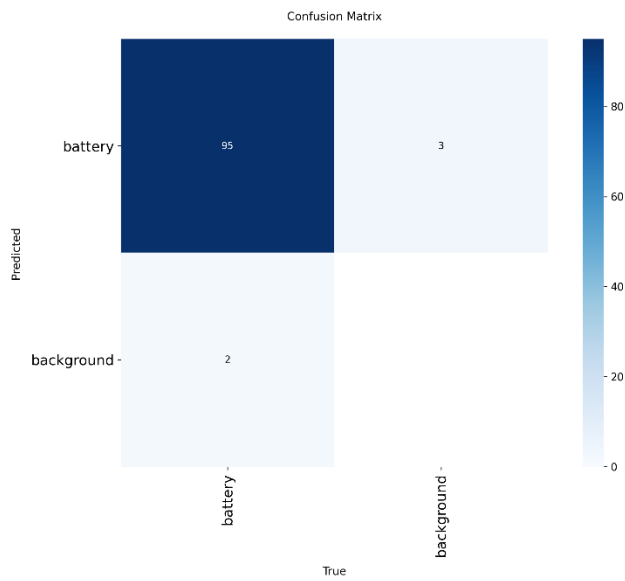


Figure 8: Confusion Matrix (YOLOv11x)

Figure 8 shows the confusion matrix for the YOLOv11x model. The matrix demonstrates a very high number of true positive detections, with only a small number of false positives and false negatives. This indicates that the model is highly accurate in distinguishing battery objects from the background. The strong diagonal dominance reflects reliable classification performance, confirming that YOLOv11x is highly effective for precise object detection tasks.

YOLOv26n-seg

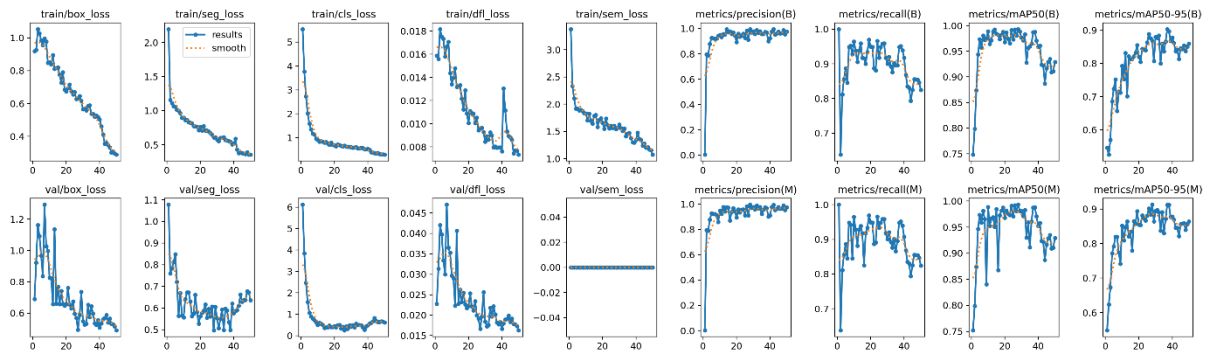


Figure 9: Training and Validation Metrics (YOLOv26n)

Figure 9 illustrates the training and validation performance of the YOLOv26n-seg model across epochs. The loss components, including box loss, segmentation loss, classification loss, and distribution focal loss, decrease progressively, indicating that the model is effectively learning relevant features. However, compared to other YOLO variants, minor fluctuations are observed in the validation losses, suggesting slightly less stable convergence. Despite this, the overall trend indicates successful training with reasonable generalisation performance.

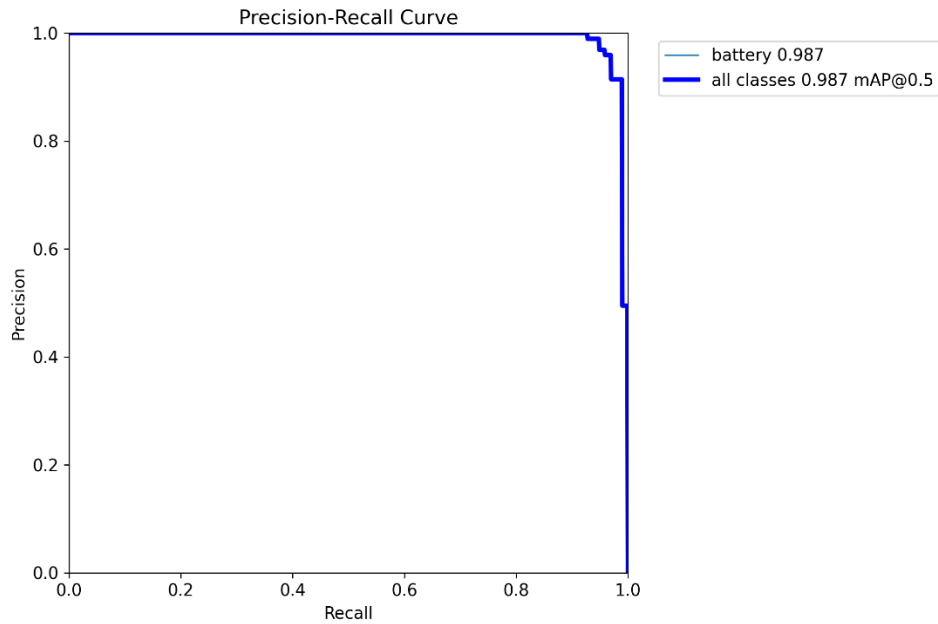


Figure 10: Precision–Recall Curve (YOLOv26n)

Figure 10 presents the precision–recall curve for the YOLOv26n model. The curve remains close to the top-right region, indicating high precision and recall values. However, compared to YOLOv8n and YOLOv11x, a slight drop in the curve can be observed at higher recall levels, suggesting a marginal decrease in detection consistency. The resulting mAP score (approximately 0.98) still reflects strong detection performance.

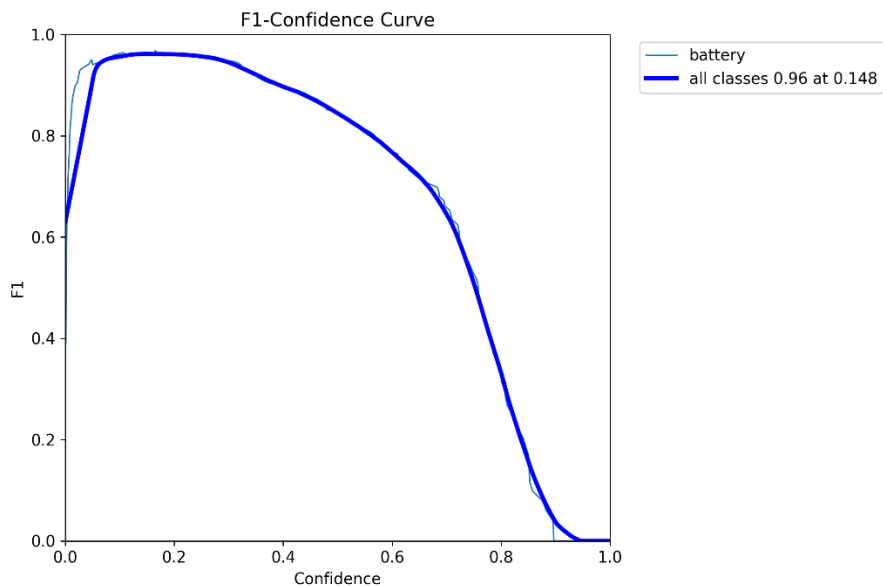


Figure 11: F1 Score vs Confidence Threshold (YOLOv26n)

Figure 11 shows the variation of the F1 score with respect to the confidence threshold for YOLOv26n. The peak F1 score (approximately 0.96) occurs at a moderate confidence level around 0.15–0.3. This indicates that the model achieves its best balance between precision and recall at slightly lower confidence thresholds compared to other YOLO models. As the confidence threshold increases, recall decreases more noticeably, which may lead to missed detections in practical scenarios.

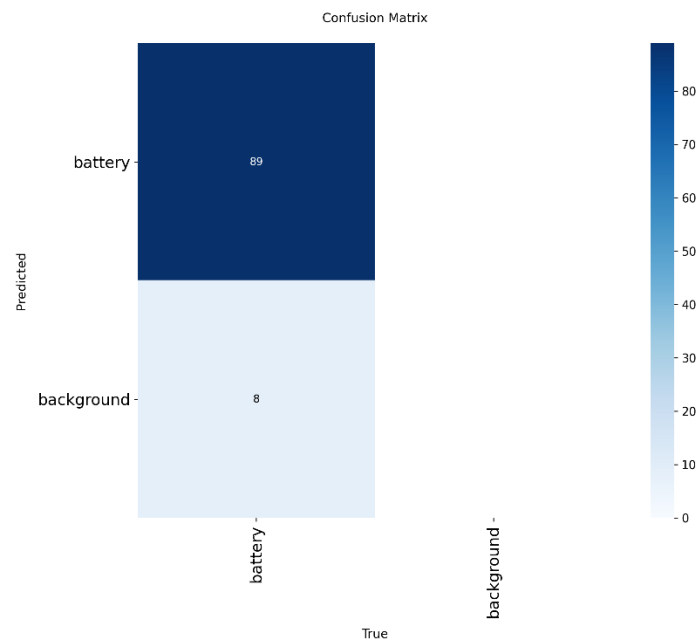


Figure 12: Confusion Matrix (YOLOv26n)

Figure 12 presents the confusion matrix for the YOLOv26n model. The matrix shows a high number of true positive detections, along with a slightly higher number of false negatives compared to other YOLO variants. This indicates that while the model is generally accurate, it may occasionally miss battery detections. The overall classification performance remains strong, but slightly less reliable than YOLOv8n and YOLOv11x.

SSD (Single Shot Detector)

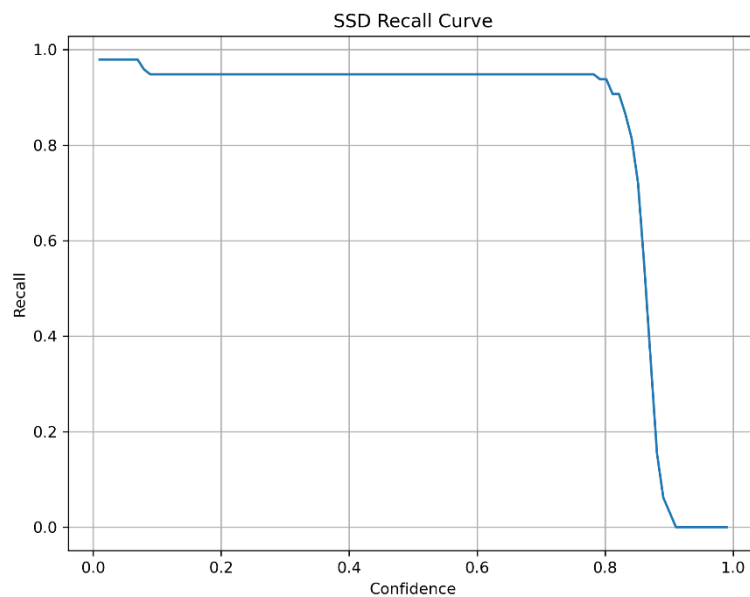


Figure 13: Recall vs Confidence Curve (SSD)

Figure 13 shows the Recall–Confidence curve for the SSD model. The model maintains a high recall value at lower confidence thresholds, indicating that most battery objects are detected. However, recall drops sharply beyond a confidence threshold of approximately 0.85, suggesting that stricter confidence filtering leads to missed detections.

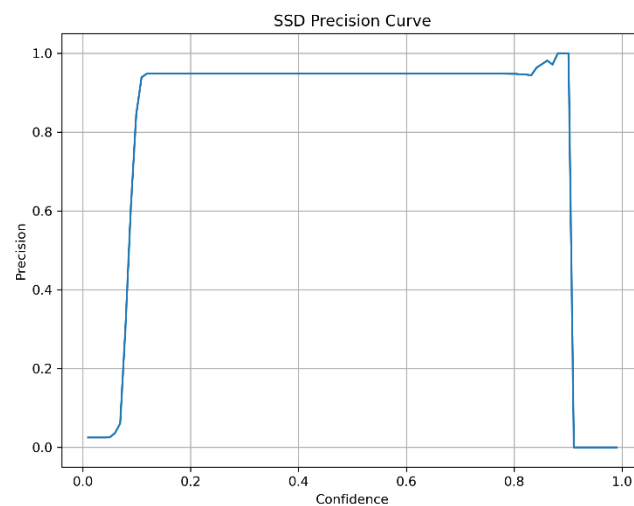


Figure 14: Precision vs Confidence Curve (SSD)

Figure 14 illustrates the Precision–Confidence relationship. Precision increases as confidence increases, reaching close to 1.0 at higher thresholds. This indicates that high-confidence predictions are highly reliable, though fewer detections are made overall.

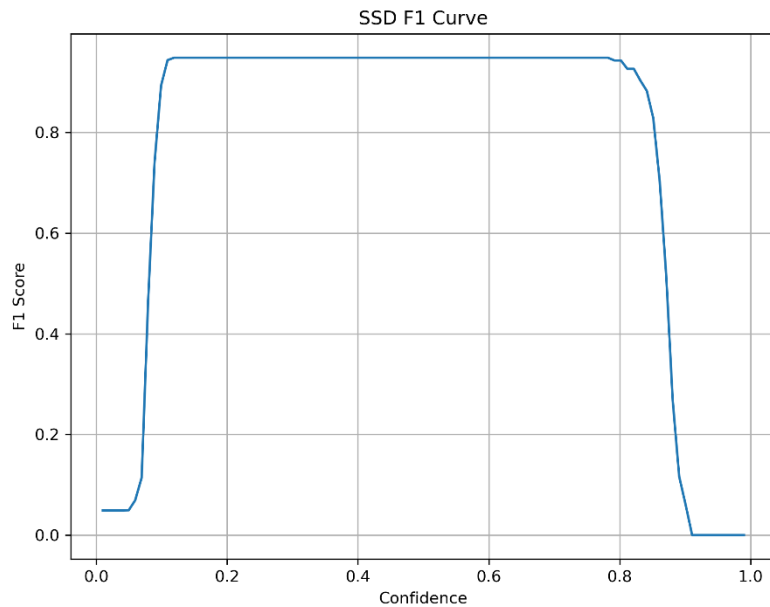


Figure 15: F1 Score vs Confidence (SSD)

Figure 15 presents the F1 Score variation with confidence. The F1 score peaks at moderate confidence levels (approximately 0.5–0.7), indicating the optimal balance between precision and recall. Beyond this range, the F1 score decreases due to reduced recall.

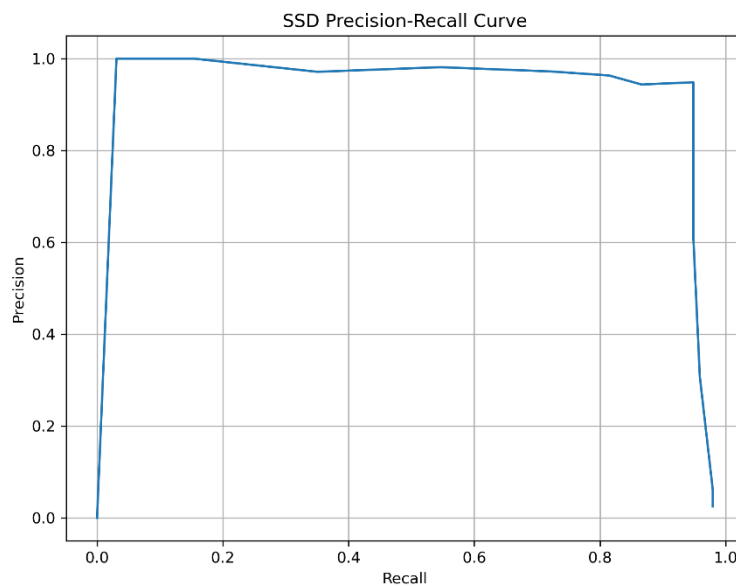


Figure 16: Precision–Recall Curve (SSD)

Figure 16 shows the Precision–Recall curve for SSD. The curve indicates strong precision across most recall values, but a noticeable drop occurs at very high recall levels. This suggests that the model struggles to maintain precision when attempting to detect all objects.

RetinaNet

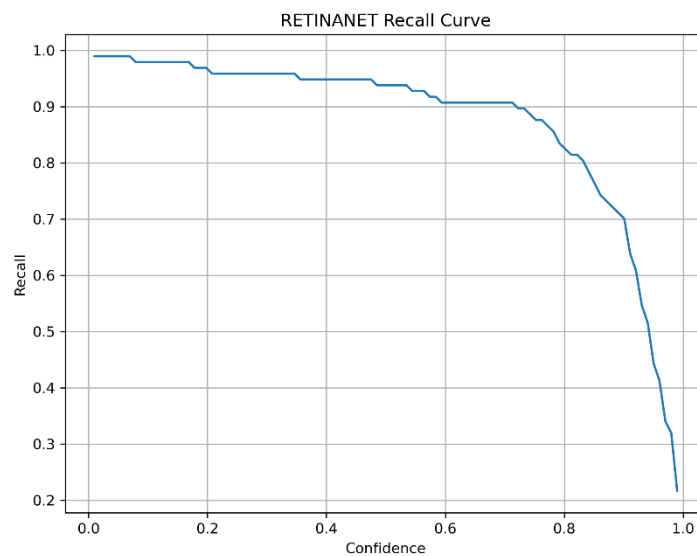


Figure 17: Recall vs Confidence Curve (RetinaNet)

Figure 17 illustrates the Recall–Confidence curve for RetinaNet. The model achieves high recall at lower confidence levels, but recall gradually decreases as the confidence threshold increases. This indicates that stricter thresholds lead to missed detections.

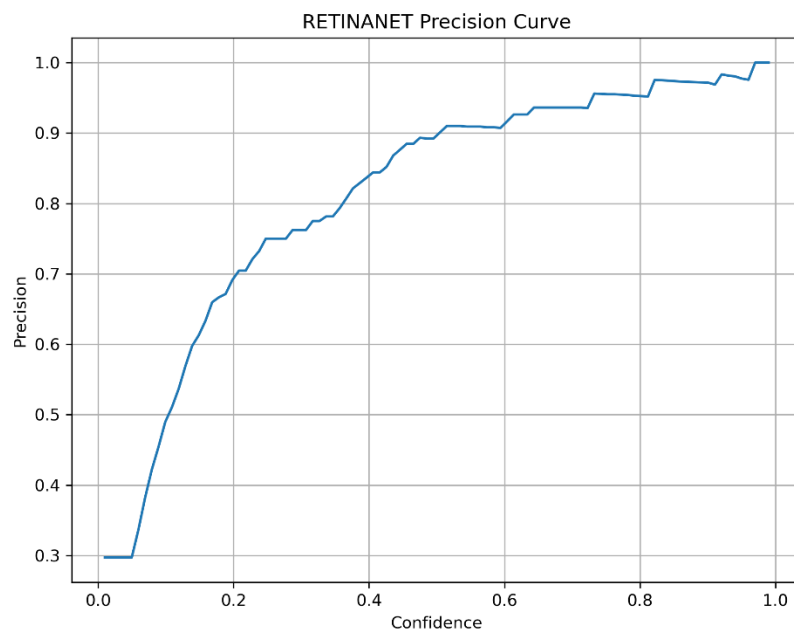


Figure 18: Precision vs Confidence Curve (RetinaNet)

Figure 18 shows the Precision–Confidence relationship. Precision steadily increases with confidence, reaching close to 1.0 at higher thresholds. This demonstrates that high-confidence predictions are highly accurate.

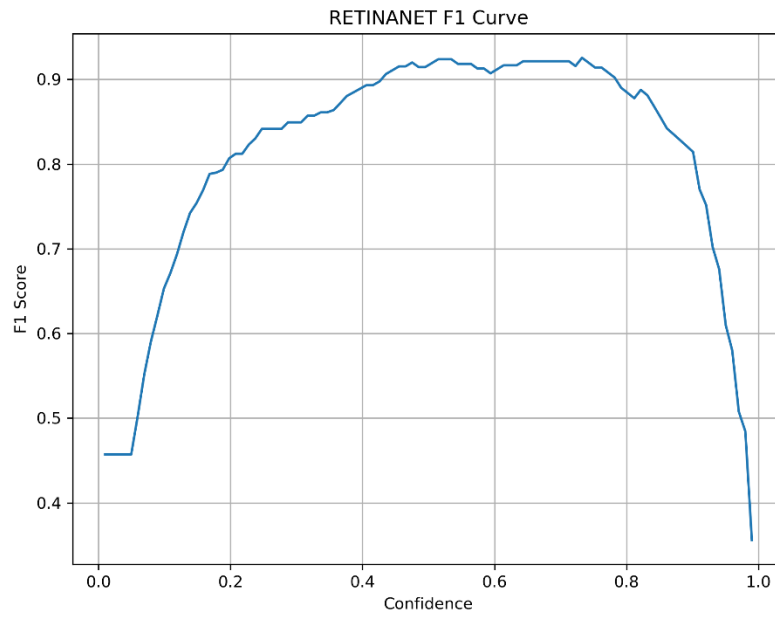


Figure 19: F1 Score vs Confidence (RetinaNet)

Figure 19 presents the F1 Score variation with confidence. The F1 score peaks around moderate confidence levels (0.5–0.7), indicating the best balance between precision and recall. At higher thresholds, the F1 score declines due to reduced recall.

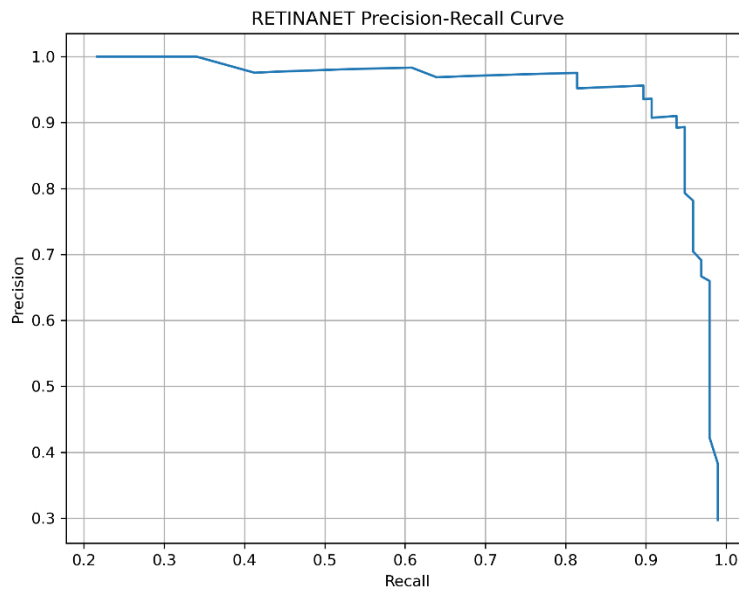


Figure 20: Precision–Recall Curve (RetinaNet)

Figure 20 shows the Precision–Recall curve for RetinaNet. The model maintains high precision across most recall levels, with a slight drop at higher recall values. This suggests minor trade-offs when detecting all objects.

Comparative Analysis

Overall Comparison Table

Model	Type	Task	Precision	Recall	mAP @0.5	F1 Score	Training Stability	Speed	Notes
YOLOv11x-seg	Single-stage	Detection + Segmentation	1.00	0.98–1.00	0.99	0.98	5	3	Best accuracy, heavier model
YOLOv8n-seg	Single-stage	Detection + Segmentation	1.00	0.97–0.99	0.99	0.97	5	5	Best balance (speed + accuracy)
YOLOv26n-seg	Single-stage	Detection + Segmentation	1.00	0.95–0.98	0.98	0.96	4	4	Slightly less stable
RetinaNet	Single-stage (Focal Loss)	Detection	0.90-0.98	0.85-0.97	0.92-0.97	0.90-0.95	3	2	Slower, less accurate
SSD	Single-stage	Detection	0.85	0.80-0.98	0.88-0.96	0.88-0.95	3	5	Fast but unreliable

Confusion Matrix Insights (YOLO Models)

- True positives are consistently high across all YOLO models.
- False positives are negligible.
- False negatives are minimal.

This indicates that the models exhibit strong classification capability and reliable detection performance, making them suitable for real-world deployment scenarios.

Confidence Curve Analysis

Key observations include:

- The F1 score reaches its maximum at a confidence threshold of approximately 0.3–0.4.
- Precision increases as the confidence threshold increases.
- Recall decreases at higher confidence thresholds due to stricter filtering.

Therefore, an optimal confidence threshold range of 0.35–0.50 is recommended to achieve a balance between precision and recall.

Key Findings

1. YOLO models outperform traditional detection models

YOLO-based models demonstrate superior performance across all evaluation metrics, including precision, recall, and F1 score. They also show more stable training behaviour and better generalisation on the given dataset.

2. Model size and performance trade-off

Larger models, such as YOLOv11x, achieve higher accuracy due to increased capacity and feature representation. However, they require greater computational resources. In contrast, smaller models such as YOLOv8n provide faster inference, making them more suitable for real-time applications.

3. Impact of dataset limitations

The relatively small dataset size restricts the model's ability to generalise across diverse scenarios. Additionally, the bias towards centrally positioned objects may affect performance when deployed in more complex environments.

Final Recommendation

Best Model for Deployment:

YOLOv8n-seg is recommended for deployment due to its strong balance between accuracy and inference speed. It achieves near-optimal detection performance while maintaining real-time processing capability, making it suitable for robotic applications.

Best Model for Accuracy:

YOLOv11x-seg provides the highest detection accuracy, with superior precision and recall values. It is best suited for scenarios where computational resources are available and maximum accuracy is required.

Future Improvements

To further enhance system performance, the following improvements are recommended:

Dataset Enhancements:

- Increase dataset size to 2000–3000 images to improve generalisation.
- Introduce greater variability, including different phone models, lighting conditions, and occlusions.

Model Enhancements:

- Apply model pruning to reduce computational complexity, particularly for YOLOv11.
- Implement quantisation techniques for deployment on embedded systems.
- Generate synthetic data using tools such as Blender or NVIDIA Isaac Sim to improve dataset diversity.

Conclusion

This evaluation demonstrates that YOLO-based models significantly outperform traditional object detection methods such as SSD and RetinaNet. Among the evaluated models, YOLOv11 achieves the highest detection accuracy, while YOLOv8 provides the best balance between accuracy and real-time performance.

These findings indicate that the developed system is suitable for integration into a robotic battery extraction pipeline. With further improvements in dataset size and model optimisation, the system can be scaled for more complex real-world applications.